

Space invaders assignment

09/30/2026 11:07:02

<https://claude.ai/chat/db6cbdc2-a9c5-407f-ba67-54342ed92cf6>

and this is the repo https://github.com/SETAPESU26/46_space-invaders d...

I'll start by pulling the repo and seeing what's broken. ``pygame.init()`` ...

did u open the repo files, shud i fork it what the fuck do i need to do tell...

Yeah, I cloned the repo and read every file (``main.py`` ,...

i cloned and installed all pip thingies, now what

Next is the BEFORE video, so don't copy anything over yet. ****1**. Run the...

i recorded before.mp4 shud i show u, its huge and idw lose credits

No, don't upload it. I don't need to see it, and I couldn't check a video...

done,

Here's the Lab-4 folder and the push. ****1**. Make the folder inside your...



Lab_4_VibeCoding_handout.pdf

and this is the repo

https://github.com/SETAPESU26/46_space-invaders

do this assignment for me

9/30/2026, 10:47:14 AM



Assessing whether using coding tools fits the assignment's rules. ▾

- 🕒 Assessing whether using coding tools fits the assignment's rules.
- ✓ Done

I'll start by pulling the repo and seeing what's broken.

Tuning enemy fire rates and difficulty settings for balance. ▾

- 📁 Clone the assigned repo and read the README
- 📁 Read all source files
- 🕒 Tuning enemy fire rates and difficulty settings for balance.
- 📁 Write fixed bullet, sounds, engine, main
- 📁 Headless tests of all four tasks
- 📁 Check what mixer format pygame.init picked
- ✓ Done

`pygame.init()` opens the mixer in stereo, so my mono buffers were being rejected. I'll make the sound generator match whatever channel count the mixer opened with.

Ran 2 commands, presented file ▾

- 📁 Fix sound channels and retest
- 📁 Package the updated code
- 📄 Presenting file(s)...
- ✓ Done

I did the code part: all four README tasks are done and tested headless. The parts that have to come from you are the videos, the push and the chat export.

What I changed

- **Task 1, collision:**

- The bug was `self.player_bullets.remove(bullet)` inside a loop over that same list, which skips the next bullet. That's why simultaneous hits got missed.
- I now build a new list of surviving bullets.
- Bullets also check collisions along the whole path they moved this frame (`swept_rect()` in `bullet.py`), so fast bullets can't skip past an enemy.
- One bullet now kills exactly one enemy.
- Test result: two bullets hitting two adjacent enemies on the same frame scored 2, and a bullet moving 60px per frame still hit.
- **Task 2, game over:** A dark overlay shows "GAME OVER" (or "YOU WIN!" if you clear the grid) and the final score, then waits for a key. The console `print` is gone. It triggers on an enemy bullet hit or enemies reaching the bottom.
- **Task 3, replay:** On the game-over screen, `R` opens a difficulty menu (`1/E` Easy, `2/M` Medium, `3/H` Hard) and `Q`/`Esc` quits. Difficulty changes the enemy speed and fire rate.
- **Task 4, sound:** Fire, enemy destroyed and game-over sounds are generated in code (`game/sounds.py`), so there are no audio files to add. If there's no audio device, the game runs silently instead of crashing.

I also lowered the enemy fire rate. The original `0.01` chance per enemy per frame comes to about 19 bullets a second with a full grid, which is nearly unwinnable. Medium is now about 1 shot a second. You can change the numbers in `DIFFICULTIES` in `game_engine.py`.

What you still need to do

1. **Videos:** Run `python main.py` on your machine, and record the "before" clip from the original repo. Then record the "after" clip from the new code. I can't record video here. Note that I tested logic headless, so give the game a quick play yourself to check it feels right.
2. **Push:** Copy the files from `Lab-4/` into your own repo, under a `Lab-4` folder. Don't raise a PR to SETAPESU26.
3. **Chat history:** Export this conversation as a PDF or doc, since the lab requires it.

Also, the lab says it's an individual assignment and that you should learn to prompt in 3-4 attempts. I'd read through the diff so you can explain it if you're asked. The core of it is the list-mutation bug, the swept collision, and the state machine (playing → game over → difficulty select).

 Lab-4_code
Zip

 game_engine
Py

 bullet
Py

 sounds
Py

 main
Py

9/30/2026, 10:48:30 AM



did u open the repo files, shud i fork it what the fuck do i need to do tell me everything

9/30/2026, 10:48:55 AM



Clarifying the fork-versus-clone workflow for pushing changes. ▾

🕒 Clarifying the fork-versus-clone workflow for pushing changes.

✓ Done

Yeah, I cloned the repo and read every file (`main.py`, `game/game_engine.py`, `player.py`, `enemy.py`, `bullet.py`, README). The fixes are built on the actual code. Here's the full process.

Fork or clone? Either works. The easiest way is to fork, since you can't push to SETAPESU26 anyway.

Step by step

1. **Fork it.** Open https://github.com/SETAPESU26/46_space-invaders and click Fork. This puts a copy in your personal GitHub account.

2. **Clone your fork.**

```
git clone https://github.com/<your-username>/46_space-invaders
cd 46_space-invaders
pip install -r requirements.txt
```

3. **Record the BEFORE video (10 sec).** Run the original code first, before touching anything:

```
python main.py
```

Screen-record (Win+G on Windows, Cmd+Shift+5 on Mac). Move, shoot, and let the game end. You'll see it just freeze with no game-over screen. Save it as

`before.mp4`.

4. **Add my code.** Download the zip from my last message, unzip it, and copy these over your clone, overwriting the old ones:

- `main.py`
- `game/game_engine.py`
- `game/bullet.py`
- `game/sounds.py` (new file)

5. **Run it and try everything:**

```
python main.py
```

Shoot (Space), get hit or let enemies reach the bottom, see the Game Over screen, press `R`, pick a difficulty, and listen for the sounds. Turn your volume up. Record 10 seconds of this as `after.mp4`.

6. **Make the Lab-4 folder.** Inside your repo, create a `Lab-4/` folder and put in:

- `before.mp4` and `after.mp4`
- the updated code (copy `main.py`, `requirements.txt`, and the `game/` folder in)
- the chat PDF (step 7)

If your instructor wants Lab-4 in a separate personal "Lab repo" instead, create that repo and put the same folder there. Check what they said if you're unsure.

7. **Export this chat.** Print the conversation to PDF (Ctrl+P / Cmd+P, then Save as PDF), or copy it into a doc. It's a required deliverable.

8. **Push:**

```
git add .  
git commit -m "Lab 4 - Space Invaders fixes and features"  
git push
```

9. **Do NOT open a Pull Request** to SETAPESU26. GitHub will show a "Contribute" button. Ignore it.

Before you submit, skim `game_engine.py` so you understand it. The main things are the list-removal bug in the collision loop (the cause of bullets passing through enemies), the swept collision in `bullet.py`, and the three game states (playing, game over, difficulty select). The lab wants you to have used an AI tool, so a chat history is expected. You should still be able to explain what changed.

If any step breaks (pygame won't install, no sound, git errors), paste the error here and I'll sort it out.

9/30/2026, 10:49:04 AM



i cloned and installed all pip thingies, now what

9/30/2026, 10:50:40 AM



Next is the BEFORE video, so don't copy anything over yet.

1. Run the original game

```
cd 46_space-invaders  
python main.py
```

If `python` doesn't work, try `python3 main.py`. A 600x700 window should open.

2. Record 10 seconds of it

- Windows: press `Win + G`, or use the Snipping Tool's record option.
- Mac: press `Cmd + Shift + 5`, then choose "Record selected portion".

While recording, move with the arrow keys or A/D and shoot with Space. Then get hit by a bullet or let the enemies reach the bottom. The game just freezes with no game-over screen, which is the broken behavior the video should show. Save the clip as

`before.mp4`.

3. Copy my code in

Unzip `Lab-4_code.zip` from my earlier message. Inside `Lab-4/`, copy these over your cloned repo, replacing the old files:

- `main.py`
- `game/game_engine.py`
- `game/bullet.py`
- `game/sounds.py` (this one is new, so it just gets added)

4. Run it again and record `after.mp4` (10 sec)

```
python main.py
```

Show these in the clip, with your volume up:

- Shoot (you'll hear a sound) and destroy some enemies (explosion sound).
- Get hit or let enemies reach the bottom. The Game Over screen shows your score and plays a sound.
- Press **R**, then pick **1**, **2** or **3** for the difficulty, and the game restarts.

Tell me once both videos are recorded and I'll walk you through the Lab-4 folder and the push. If the game doesn't open or crashes, paste the error.

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i recorded before.mp4 shud i show u, its huge and idw lose credits

9/30/2026, 10:53:09 AM



No, don't upload it. I don't need to see it, and I couldn't check a video against your code anyway. Just keep the file safe on your computer.

Big video files are fine for GitHub as long as each one is under 100 MB. If `before.mp4` is bigger than that, GitHub will reject the push. In that case, compress it (HandBrake is free) or trim it to about 10 seconds.

Next steps

1. Copy my updated files into your cloned repo, replacing the old ones:
 - `main.py`
 - `game/game_engine.py`
 - `game/bullet.py`
 - `game/sounds.py` (new file)
2. Run `python main.py` and try everything: shoot, kill enemies, get a Game Over, press `R`, and pick a difficulty. Record 10 seconds of that as `after.mp4`.
3. Tell me when that's done and I'll walk you through the Lab-4 folder and the push.

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done,

9/30/2026, 10:59:23 AM



Here's the Lab-4 folder and the push.

1. Make the folder inside your cloned repo

```
cd 46_space-invaders
mkdir Lab-4
```

2. Put these in `Lab-4/`

- `before.mp4` and `after.mp4`
- the updated code: copy `main.py`, `requirements.txt` and the whole `game/` folder in
- the chat PDF (see step 3)

Your repo should end up looking like this:

```
46_space-invaders/
├── main.py, game/, ...      (updated code, already there)
└── Lab-4/
    ├── before.mp4
    ├── after.mp4
    ├── main.py
    ├── requirements.txt
    ├── game/
    └── Chat_History.pdf
```

3. Export this chat

Open the conversation, press `Ctrl+P` (or `Cmd+P` on Mac), choose "Save as PDF", and save it as `Chat_History.pdf` in `Lab-4/`. Do this last so the PDF includes everything.

4. Check file sizes

GitHub rejects any file over 100 MB. Check with `ls -lh Lab-4` (Mac/Linux) or look at the file properties on Windows. If a video is too big, compress or trim it first.

5. Push

```
git add .  
git commit -m "Lab 4 - Space Invaders fixes and features"  
git push
```

It may ask you to log in. If it wants a password, GitHub needs a personal access token or the GitHub Desktop login instead of your account password.

6. Verify

Open your fork on github.com and click into `Lab-4/`. Make sure both videos, the code and the PDF are there. Click a video to check that it uploaded and plays.

7. Don't raise a PR. Ignore any "Contribute" or "Open pull request" button.

Depending on your instructor's wording, they may want the submission link to be your repo's `Lab-4` folder, so copy that URL when you're done. If you hit any error on the push, paste it here.

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